European Commission

SCIENCE FOR POLICY BRIEFS

Towards the Implementation of an EU Strategy for Technology Infrastructures

Headlines

- **Develop an EU Strategy for Technology Infrastructures (TIs) using the European Strategy for Research Infrastructures (ESFRI) as a source of inspiration when relevant, and designing dedicated processes adapted to TIs' specificities.**
- **Set up an agile governance of TIs at EU level including Member States' experts responsible for TIs, and stakeholders including TIs managers, and public and private users.**
- **Develop clear pathways for public support of capital and operating expenditures (CAPEX and OPEX), supporting the long-term sustainability of TIs.**
- **Combine and complete existing TIs' mappings and use existing EU instruments for roadmaping the future needs for TIs, e.g. European Partnerships, ERA Industrial Technology Roadmaps, Important Projects of Common European Interest (IPCEIs), Industrial Alliances, etc.**
- **Put in place investments' prioritisation mechanisms at EU level for European-scale TIs and synchronisation of national/regional investment plans for regional-scale TIs.**
- **Foster pan-European accessibility to TIs by strengthening their use in competitively funded projects at EU, national and regional levels, defining standardised principles for access, etc.**

Background

Technology Infrastructures (TIs) are (physical or virtual) facilities and equipment such as demonstrators, testbeds, piloting facilities, living labs. They are used to develop, mature, test, demonstrate and upscale technology to advance through industrial research and experimental development activities from proof of concept to technology validation in relevant environment.

TIs are mostly managed and hosted by Research Performing Organisations - mainly Research and Technology Organisations (RTOs) and Technical Universities (TUs), which collaborate to jointly develop and integrate innovative technologies into new products, processes and services. TIs managers have the key role to create, operate, maintain, upgrade, and decommission TIs, requiring very high and sustainable investments that industry cannot afford.

TIs are open to public and private users including large companies, SMEs and startups, and they play a key role for industry's uptake of technology. TIs are enablers for technology transfer from basic research to the market, especially to SMEs, and for regional development, as they are embedded in Smart Specialisation Strategies.

Research Infrastructures (RIs) and TIs are complementary. RIs create new knowledge which will be key for TIs to address the future needs of European Industry. The same facility can play both roles, depending for example on the technology readiness level (TRL) of the projects carried out. An RI could also evolve into a TI as the knowledge and technology in a specific sector matures.

Overall, TIs are essential to boost industrial competitiveness and to deliver on the green and digital transitions, thus calling for a high-level dynamism and improved quality of TIs.

Policy Context

The strategic importance of TIs has not sufficiently been recognised yet in European, national and regional policies in Europe, with a few exceptions.At the EU level, TIs have been the focus of a limited number of EC policy documents and workshops over the past decade.

The JRC has been building key expertise on TIs thanks to the European Technology Transfer Offices' (TTO) Circle. Together with RISE and Teknikföretagen, the JRC organised a workshop on best practices in the opening up of RTOs' TIs to industry for testing, demonstration and co-creation (2017). The JRC has also set up a programme to open up its own research infrastructures to a wide range of users from the public and private sectors. Some of the main challenges faced in opening up these infrastructures are linked to the management of intellectual property rights (IPR) and the ownership of the data created, as well as financial considerations with regards to the amount to be charged to

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users, and finding the right balance between accessibility and sustainability.

On the side of DG RTD, both the Action Plan to Promote the access of SMEs to KETs Technology Infrastructures (2015) and the Staff Working Document (SWD) on Technology Infrastructures (2019) recognise TIs as essential components of RD&I ecosystems, and crucial to foster the uptake of Europe's research results by the market. The four main challenges that have been underlined in the SWD are:

- Visibility: it is not always easy for users to find and access the services and facilities they need.
- Prioritisation: the granularity and targeted actions between the EU, national and regional levels need to be streamlined.
- Accessibility: quick and easy access to TIs is essential for companies of all sizes, but there are some barriers to overcome (access conditions, IPR, cost, etc.).
- Networks: connecting TIs together and to relevant RIs following a value-chain driven approach is key to sharing competences and best practices.

More recently, the EU Industrial Strategy (March 2020) points to two challenges: (1) strategic autonomy – the EU needs to become more independent in strategic areas for instance from China and the US, and (2) delivering the twin transitions towards green and digital. The role of TIs will be essential to achieve these two objectives, and a strong connection between RD&I and industrial policy is essential for a coherent European approach to these overarching EU objectives.

TIs have also been recognised as a key element of the new European Research Area (ERA). The EC Communication "A new ERA for Research and Innovation" (Sept. 2020) stresses that "*Industry, and notably SMEs, require access to the right technology infrastructures to quickly develop and test their innovations and successfully enter the market.*". This EC Communication has been endorsed by the Council Conclusions on the New ERA (Dec. 2020), which calls for the development of an EU Strategy for Technology Infrastructures.

The strategic importance of TIs in RD&I ecosystems has also been highlighted in the EC Proposal for a Council Recommendation on a Pact for Research and Innovation in Europe (July 2021). This was the basis for the Council Recommendations on a Pact for R&I in Europe, which includes a concrete Policy Agenda for the new ERA with a dedicated action plan for the next three years.

Methodology and Focus

In this context, the JRC and the European Association of Research and Technology Organisations (EARTO) launched a project on TIs to gather evidence and highlight the specificities of TIs, assess the challenges they face over their whole lifecycle from inception to decommissioning, and identify how their capacity could be further leveraged accross Europe.

In addition to the review of the state of the art for TIs, the methodology used to collect input for this project includes:

- A questionnaire to collect input from TI managers with 18 responses received (September-October 2021).
- A joint JRC-EARTO workshop with nearly 50 experts on TIs (30 September 2021).
- EARTO Policy Event focusing on the new EU Strategy for TIs to deliver the twin transitions (27 October 2021).
- CONCORDi conference EARTO-JRC session on Technology Infrastructures (22-25 November 2021).

This input led to considerations on how to take stock of the work carried out so far with the European Strategy for Research Infrastructures (ESFRI) to design a dedicated process adapted to the specificities of TIs (e.g. value-chain perspective, different types of RD&I actors to be involved in the road mapping and gap analysis), while preserving an effective connection between ESFRI and the new European strategy on TIs without creating extra burden.

This joint project by the JRC and EARTO has resulted in policy recommendations to support the implementation of the ERA Policy Agenda with a concrete action plan to develop an integrated European TIs landscape. These recommendations are presented in the following paragraphs structured in three parts: the role of the TIs in the RD&I ecosystem; ensuring TIs long-term sustainability; and towards an EU strategy for TIs.

The Role of TIs in the RD&I Ecosystem

TIs are a condition for the emergence and success of RD&I ecosystems: they are a key instrument to bring together public research, SMEs, large companies, and organise technology transfer from laboratories to industry innovations. TIs cannot be considered as an isolated asset: they need to be connected to other TIs and RIs across different sectors and disciplines at European, national, and regional levels:

- TIs are essential to the well-functioning of efficient European RD&I ecosystems. A better integration of TIs in policy initiatives at European, national and regional levels could further leverage their potential. A Pilot Board led by a Board Chair should be established to support the establishment of the European Strategy for TIs and increase its visibility.
- TIs are not just physical or digital infrastructures, they also require interdisciplinary competences and complex technological know-how and experts to operate them. An alignment with the European Skills Agenda would enable to avoid gaps.
- The TI landscape is very diverse. Streamlining and analysing the existing European mappings of TIs would enable to: better understand their specificities, refine and create further typologies of TIs, and achieve better tailor-made, evidence-based policies and investments.
- RIs and TIs have each their own specificities. The EU Strategy for TIs could draw inspiration from ESFRI processes and modus operandi when relevant, and design dedicated ones when needed - e.g. different stakeholders involved, bottom-up and value-chain approach for roadmapping of needs, different pace to keep up to speed with technology development, etc.
- RIs and TIs need to be taken as part of an ecosystem and not per opposition or disconnected. There is a whole continuum between RIs and TIs, with expertise and associated services which are often complementary. Links

between ESFRI and the future EU Strategy for TIs need to be built from the onset.

Ensuring TIs long-term Sustainability

Each process to build a TI is quite unique, as different business sectors or research fields have very different needs. However, they all share some common factors, which are essential for the success of a new TI, which include the following:

- TIs managers' foresight capabilities should be further leveraged in dedicated policy, i.e. understanding where the research forefront is going and identifying the future needs of industry.
- The future users (public & private) of a TI need to be involved throughout the TI's creation process to ensure its long-term use.
- A sound business model ensuring the long-term sustainability of the TI is essential from the onset of a TI's creation process. TIs need to be upgraded at the same pace than the technologies and the products that are developed and tested: fast. More certainty on public funding cycles would leverage TIs' capabilities.
- Clear pathways of grant-based public support for capital investments for TIs (CAPEX) adapted to the speed of technology development is essential, as the funding landscape is very scattered. Creating synergies for a more structural support to TIs at European, national and regional levels would be of high added value.
- There is a lack of public investment plans focusing on the upgrade of existing TIs at European and national levels. These should be designed in complementarity with public investments targeting the creation of new TIs. It is important to have a balanced approach (creating versus upgrading): not keeping existing TIs at the forefront of innovation may result in a suboptimal use of public resources.

Towards an EU Strategy for TIs

The EC SWD on TIs suggests that "there is a critical momentum for the EU together with Member States to be more ambitious, exploring with relevant national and regional stakeholders a shared vision and jointly developing a European strategy for technology infrastructures to support industry scale-up and technology diffusion across Europe". This recommendation has recently been taken up in the frame of the new ERA, with an EC Communication and Council Conclusions including targeted policy actions for TIs. To ensure the sound implementation of these policy actions, the following recommendations are brought up:

- Taking stock of the public policy initiatives and instruments, that currently exist to support TIs at regional, national and European levels via a systematic policy observatory, would provide opportunities for benchmarking and streamlining investments.
- Combining and completing the existing repositories and mappings of TIs at EU level covering both TIs' locations and the services and facilities they offer could be used to (1) enable a better understanding of the TIs' landscape by policy-makers and facilitate gap analyses; (2) foster accessibility to TIs (if coupled with an efficient marketing campaign to increase visibility); and (3) create connections between complementary TIs.
- Roadmapping of future needs for CAPEX investments in TIs should be organised (1) with a sectorial value-chain and bottom-up approach, (2) with the involvement of TIs stakeholders (managers and users), (3) by identifying the future needs for TIs in existing roadmaps linked to current EU instruments and actions (e.g. European Partnerships, ERA Industrial Technology Roadmaps, Industrial Alliances, IPCEIs, etc.).
- Setting up a mechanism to draw from sectorial roadmaps and prioritise and/or coordinate investments in TIs at European level in strategic sectors would be valuable to maximise the use of public funds.
- Creating an agile advisory board will be necessary to operationalise the EU Strategy on TIs (incl. investments prioritisation and coordination). It should be composed of experts responsible for TIs within national ministries, as well as TI managers (foresight) and users (long-term use). The Advisory Board should be framed within an overarching governance structure and have a fixed term duration to adapt to the evolution of the RD&I ecosystem.
- The prioritisation mechanism and coordination of national/regional TIs' roadmaps needs to take into account the different types of TIs: (1) for TIs of European scale and market niche, duplication should be avoided to achieve economies of scale and a broader user base; whereas (2) for TIs of regional focus, replication at regional level is key to ensure accessibility and avoid gaps to bridge the innovation divide.
- The pan-European accessibility to TIs should be fostered by strengthening their use in competitively funded projects, defining standardised principles for access to TIs which could be implemented on a voluntary basis (incl. IPR, Data management, etc.) and investing in the digitalisation of TI processes when possible.
- Creating thematic networks of TIs with a value-chain approach at EU level would enable to: better integrate and structure the European landscape for TIs and increase the synergies with RIs; foster capacity building across regions; and spread excellence and expertise to overcome the European innovation divide.

Next steps

The present Policy Brief provides recommendations drawn from TI managers and users to support the implementation of the ERA Action Plan towards an Integrated European landscape for TIs over the next three years.

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